

Summary Report: How Road Infrastructure Affects Accident Rates

Road infrastructure greatly influences traffic safety. This report analyzes accident data to identify high-risk road features. The findings show that Traffic Signals (36.22%), Crossings (26.89%), and Junctions (17.33%) account for most accidents, making them critical danger zones.

Other features like Stop Signs (6.48%) and Stations (6.05%) contribute to accidents but at lower rates. In contrast, roundabouts, bumps, and turning loops show minimal accidents, suggesting their safety benefits.

The high accident rates at Traffic Signals, Crossings, and Junctions stem from high traffic conflict points, complex driver decisions, and congestion-related risks. To improve safety, urban planners should:

- Enhance intersection design with better management and visibility.
- Optimize traffic signals to ease congestion and prevent red-light running.
- Improve pedestrian crossings with better lighting and raised crosswalks.
- Simplify junction layouts or replace them with roundabouts where feasible.
- Increase enforcement at high-risk areas to deter unsafe driving.

By focusing on these improvements, accident rates can be reduced, creating safer roads for all users. Further research into accident types at these locations would refine safety strategies.